

Properties of the Integral

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In the discussion that follows:

- Let $(\Omega, \mathcal{F}, \mu)$ be a measure space.

- The notation $I(f)$ and $\int f d\mu$ is used interchangeably.

1. Let f, g be nonnegative integrable functions on Ω such that $g \leq f$. Then, $I(g) \leq I(f)$.

Proof. Let $f_n \uparrow f$ and $g_n \uparrow g$, where f_n and g_n are monotonically increasing sequences of simple functions. Define $h_n = \min\{f_n, g_n\}$. We first show that h_n is monotonically increasing, i.e., $h_n \leq h_{n+1}$ for all $n \in \mathbb{N}$.

Suppose $h_n = f_n$, i.e., $f_n \leq g_n$. If $h_{n+1} = f_{n+1}$, then $h_n \leq h_{n+1}$ since f_n is monotonically increasing. On the other hand, if $h_{n+1} = g_{n+1}$, then $g_{n+1} \geq g_n \geq f_n = h_n$. Similarly, if $h_n = g_n$, then also $h_n \leq h_{n+1}$, and so h_n is monotonically increasing.

Since $g \leq f$, for a large enough n , $g_n \leq f_n$. Hence, $\lim_{n \rightarrow \infty} h_n = \lim_{n \rightarrow \infty} g_n = g$ and so $h_n \uparrow g$. Also, by the way h_n is defined, $h_n \leq f_n$ for all $n \in \mathbb{N}$, and so $I(h_n) \leq I(f_n)$, since h_n and f_n are simple functions. Therefore, $I(g) = \lim_{n \rightarrow \infty} I(h_n) \leq \lim_{n \rightarrow \infty} I(f_n) = I(f)$. **QED**

2. Let $f : \Omega \rightarrow \overline{\mathbb{R}}$ be an integrable function and $A \in \mathcal{F}$. Then, $\chi_A f$ is also integrable, and we denote $I(\chi_A f)$ as $\int_A f d\mu$.

Proof. Since $f^+ = \max\{0, f\}$ and since $\chi_A \in \{0, 1\}$, $(\chi_A f)^+ = \chi_A f^+ \leq f^+$. By Property 1, $I((\chi_A f)^+) \leq I(f^+) < \infty$ as f is integrable. Hence, $(\chi_A f)^+$ is integrable. Also, $f^- = \max\{0, -f\}$ and so, $(\chi_A f)^- = \chi_A f^- \leq f^-$. Once again applying Property 1, we get $I((\chi_A f)^-) \leq I(f^-) < \infty$, and thus, $(\chi_A f)^-$ is also integrable. Therefore, $\chi_A f$ is integrable. **QED**

3. Let $f, g : \Omega \rightarrow \overline{\mathbb{R}}$ be nonnegative integrable functions. Then, $I(f + g) = I(f) + I(g)$ and $I(cf) = cI(f)$ for $c \geq 0$.

Proof. Let $f_n \uparrow f$ and $g_n \uparrow g$, where f_n, g_n are sequences of monotonically increasing, non-negative simple functions. Then, $f_n + g_n \uparrow f + g$ and $f_n + g_n$ is a simple function for all $n \in \mathbb{N}$. Hence, $I(f + g) = \lim_{n \rightarrow \infty} I(f_n + g_n) = \lim_{n \rightarrow \infty} (I(f_n) + I(g_n))$, by the linearity of the integral over simple functions. Therefore, $I(f + g) = \lim_{n \rightarrow \infty} I(f_n) + \lim_{n \rightarrow \infty} I(g_n) = I(f) + I(g)$.

Also, $f_n \uparrow f$ implies that $cf_n \uparrow cf$ and since cf_n is also a monotonically increasing sequence of nonnegative simple functions (for $c \geq 0$), $I(cf) = \lim_{n \rightarrow \infty} I(cf_n) = \lim_{n \rightarrow \infty} cI(f_n) = cI(f)$, as desired. **QED**

4. Let $A, B \in \mathcal{F}$ be disjoint and $f : \Omega \rightarrow \overline{\mathbb{R}}$ be integrable. Then, $\int_{A \cup B} f d\mu = \int_A f d\mu + \int_B f d\mu$.

Proof. We first show that if f is integrable over A and B , then it is also integrable over $A \cup B$. That is, $\chi_A f$ and $\chi_B f$ are integrable, and we want to show that $\chi_{A \cup B} f$ is also integrable.

$(\chi_{A \cup B} f)^+ = \chi_{A \cup B} f^+ = (\chi_A + \chi_B) f^+ = \chi_A f^+ + \chi_B f^+$. By Property 3, $I((\chi_{A \cup B} f)^+) = I(\chi_A f^+) + I(\chi_B f^+)$, and since f is integrable over A and B , $I(\chi_A f^+)$ and $I(\chi_B f^+)$ are both finite, and thus their sum is finite. Hence, $I((\chi_{A \cup B} f)^+)$ is finite. Similarly, $I((\chi_{A \cup B} f)^-)$ is also finite, and therefore $\chi_{A \cup B} f$ is integrable.

$\int_{A \cup B} f d\mu = \int \chi_{A \cup B} f d\mu$ and since A and B are disjoint,

$$\begin{aligned} \int \chi_{A \cup B} f d\mu &= \int (\chi_{A \cup B} f)^+ d\mu - \int (\chi_{A \cup B} f)^- d\mu \\ &= \int \chi_A f^+ d\mu + \int \chi_B f^+ d\mu - \int \chi_A f^- d\mu - \int \chi_B f^- d\mu \quad (\text{by Property 3}) \\ &= \left(\int \chi_A f^+ d\mu - \int \chi_A f^- d\mu \right) + \left(\int \chi_B f^+ d\mu - \int \chi_B f^- d\mu \right) \\ &= \int_A f d\mu + \int_B f d\mu. \end{aligned}$$

QED

5. If $f : \Omega \rightarrow \overline{\mathbb{R}}$ is integrable, then $|f| < \infty$ a.e.

Proof. f is integrable. We want to show that $|f| < \infty$ a.e., that is, if $E \in \mathcal{F}$ is such that $|f(x)| = \infty$ for all $x \in E$, then $\mu(E) = 0$.

Since $|f| = f^+ + f^-$, by Property 3 we have that $I(|f|) = I(f^+) + I(f^-)$, and since $I(f^+), I(f^-) < \infty$ as f is integrable, $I(|f|)$ is also finite and therefore $|f|$ is integrable.

We first note that

$$\int_{\{|f(x)| \geq n\}} |f| d\mu \leq \int |f| d\mu < \infty$$

and also that

$$\int_{\{|f(x)| \geq n\}} n d\mu \leq \int_{\{|f(x)| \geq n\}} |f| d\mu.$$

Hence,

$$n\mu(\{|f(x)| \geq n\}) \leq \int_{\{|f(x)| \geq n\}} |f| d\mu,$$

i.e.,

$$\mu(\{|f(x)| \geq n\}) \leq \frac{1}{n} \int_{\{|f(x)| \geq n\}} |f| d\mu.$$

Now, define $A_n := \{x : |f(x)| \geq n\}$. Since f is integrable, $A_n \in \mathcal{F}$. Clearly, $A_{n+1} \subseteq A_n$. Define $A := \bigcap_{n \geq 1} A_n$. Then, $A \in \mathcal{F}$. Define $E := \{x : |f(x)| = \infty\}$. We now show that $E = A$.

Let $x \in A$. Then, $x \in A_n$ and therefore $|f(x)| \geq n$ for all $n \in \mathbb{Z}$. Hence, $|f(x)| = \infty$ and so $x \in E$. Thus, $A \subseteq E$. Now suppose $x \in E$. Then, $|f(x)| = \infty$ and so $|f(x)| \geq n$ for all $n \in \mathbb{N}$, implying that $x \in A$. Thus, $E \subseteq A$ and therefore $A = E$.

Since $\mu(A_1)$ is finite, we can apply continuity from above to obtain

$$\mu(E) = \lim_{n \rightarrow \infty} \mu(A_n) \leq \lim_{n \rightarrow \infty} \frac{1}{n} \int_{A_n} |f| d\mu \leq \lim_{n \rightarrow \infty} \frac{1}{n} \int |f| d\mu = 0$$

as desired. QED

6. If $f, g : \Omega \rightarrow \overline{\mathbb{R}}$ are nonnegative and integrable, then $\int (f - g) d\mu = \int f d\mu - \int g d\mu$.

Proof. Define $h := f - g$. We assume that h is well-defined, that is, f and g are not $+\infty$ simultaneously. Then, $h = h^+ - h^- = f - g$. If h is finite, then $h^+ + g = h^- + f$. If $h = +\infty$, then $f = +\infty$, g is finite, $h^+ = +\infty$, and h^- is finite. Therefore, $h^+ + g = +\infty = h^- + f$. Similarly, $h^+ + g = h^- + f$ holds even when $h = -\infty$. Hence, $\int (h^+ + g) d\mu = \int (h^- + f) d\mu$. Applying Property 3 gives $\int h^+ d\mu + \int g d\mu = \int h^- d\mu + \int f d\mu$, i.e., $\int h^+ d\mu - \int h^- d\mu = \int f d\mu - \int g d\mu$, and since $\int h d\mu = \int h^+ d\mu - \int h^- d\mu$, we have the desired result. QED

7. If $f, g : \Omega \rightarrow \overline{\mathbb{R}}$ are integrable, then $f + g$ is also integrable and $\int (f + g) d\mu = \int f d\mu + \int g d\mu$.

Proof. We first show that $f + g$ is integrable. $(f + g)^+ \leq f^+ + g^+$ and therefore, by Properties 1 and 3, $I((f + g)^+) \leq I(f^+) + I(g^+) < \infty$ since $I(f^+), I(g^+) < \infty$, as f, g are integrable. Similarly, $I((f + g)^-) < \infty$ and so $f + g$ is integrable.

Now, $f + g = (f^+ - f^-) + (g^+ - g^-) = (f^+ + g^+) - (f^- + g^-)$. By Property 5,

$$\begin{aligned} \int (f + g) d\mu &= \int (f^+ + g^+) d\mu - \int (f^- + g^-) d\mu \\ &= \left(\int f^+ d\mu + \int g^+ d\mu \right) - \left(\int f^- d\mu + \int g^- d\mu \right) \quad (\text{by Property 3}) \\ &= \left(\int f^+ d\mu - \int f^- d\mu \right) + \left(\int g^+ d\mu - \int g^- d\mu \right) \\ &= \int f d\mu + \int g d\mu. \end{aligned}$$

QED

8. If $f : \Omega \rightarrow \overline{\mathbb{R}}$ is integrable, then $|\int f d\mu| \leq \int |f| d\mu$.

Proof. We have,

$$\begin{aligned} \left| \int f d\mu \right| &= \left| \int f^+ d\mu - \int f^- d\mu \right| \\ &\leq \int f^+ d\mu + \int f^- d\mu \\ &= \int (f^+ + f^-) d\mu = \int |f| d\mu. \end{aligned}$$

QED

9. Let $f : \Omega \rightarrow \overline{\mathbb{R}}$ be integrable and $c \in \mathbb{R}$. Then, cf is integrable and $\int cf d\mu = c \int f d\mu$.

Proof. $\int |cf|d\mu = \int |c||f|d\mu = |c| \int |f|d\mu$, where the last equality is true by Property 3. $\int |f|d\mu = \int (f^+ + f^-)d\mu = \int f^+d\mu + \int f^-d\mu < \infty$ as f is integrable, and so $\int |cf|d\mu < \infty$. Since $\int (cf)^+d\mu$ and $\int (cf)^-d\mu$ are both bounded by $\int |cf|d\mu$, they are also finite and hence cf is integrable.

If $c = 0$, the statement is trivial.

If $c > 0$,

$$\begin{aligned} \int cf d\mu &= \int (cf)^+ d\mu - \int (cf)^- d\mu \\ &= \int cf^+ d\mu - \int cf^- d\mu \\ &= c \left(\int f^+ d\mu - \int f^- d\mu \right) \\ &= c \int f d\mu. \end{aligned}$$

Further, if $c < 0$,

$$\begin{aligned} \int cf d\mu &= \int (cf)^+ d\mu - \int (cf)^- d\mu \\ &= \int (-c)f^- d\mu - \int (-c)f^+ d\mu \\ &= c \int f d\mu. \end{aligned}$$

QED

10. If $f \geq 0$, then $\int f d\mu \geq 0$.

Proof.

$$\begin{aligned} \int f d\mu &= \int f^+ d\mu - \int f^- d\mu \\ &= \int f^+ d\mu - 0 \geq 0. \end{aligned}$$

QED

11. If $f \geq g$, then $\int f d\mu \geq \int g d\mu$.

Proof. Since $f \geq g$, $f - g \geq 0$. By Property 10, $\int (f - g) d\mu \geq 0$, and by linearity, $\int f d\mu - \int g d\mu \geq 0$, i.e., $\int f d\mu \geq \int g d\mu$. **QED**

12. If $f \geq 0$ and $\int f d\mu = 0$, then $f = 0$ a.e.

Proof. Define $E_n := \{x : f(x) \geq \frac{1}{n}\}$. Let $E = \bigcup_{n \geq 1} E_n$ and $F = \{x : f(x) > 0\}$. We first show that $E = F$. Let $x \in E$. Then, for some $n \geq 1$, $x \in E_n$, i.e., $f(x) \geq \frac{1}{n}$, and since $n \in \mathbb{N}$, $f(x) > 0$. Hence, $x \in F$ and $E \subseteq F$. Also, if $x \in F$, then $f(x) > 0$, and so there

exists $n \in \mathbb{N}$ such that $f(x) \geq \frac{1}{n}$, i.e., $x \in E_n$. Therefore, $F \subseteq E$, showing that $E = F$. Further, $E_n \subseteq E_{n+1}$ and so $E_n \uparrow E$.

Now,

$$\int_{E_n} \frac{1}{n} d\mu \leq \int_{E_n} f d\mu \leq \int f d\mu = 0$$

and so $\int_{E_n} \frac{1}{n} d\mu = 0$. Hence, $\frac{1}{n} \int \chi_{E_n} d\mu = 0$, i.e., $\mu(E_n) = 0$, for all $n \in \mathbb{N}$. By the subadditivity of μ , $\mu(E) \leq \sum_{n \geq 1} \mu(E_n) = 0$. Since $E^c = \{x : f(x) = 0\}$ (as $f \geq 0$), and since $\mu(E) = 0$, $f = 0$ a.e. **QED**

13. If $f = g$ a.e., then $\int f d\mu = \int g d\mu$.

Proof. $f = g$ a.e., i.e., $f(x) = g(x)$ for all $x \in E$ and $\mu(E^c) = 0$. $\int f d\mu = \int (\chi_E + \chi_{E^c}) f d\mu = \int \chi_E f d\mu + \int \chi_{E^c} f d\mu$. Since $\mu(E^c) = 0$, $\int \chi_{E^c} f d\mu = 0$, and so $\int f d\mu = \int \chi_E f d\mu = \int \chi_E g d\mu$, as $f = g$ over E . Also, $\int \chi_{E^c} g d\mu = 0$ and so $\int \chi_E g d\mu = \int \chi_E g d\mu + \int \chi_{E^c} g d\mu = \int g d\mu$. Therefore, $\int f d\mu = \int g d\mu$. **QED**

14. Let $h : \Omega \rightarrow \overline{\mathbb{R}}$ be measurable and $|h| \leq f$, where f is integrable. Then, h is integrable.

Proof. Since $|h| \leq f$, $h^+ + h^- \leq f$. By Property 11, $I(h^+ + h^-) \leq I(f)$, and by Property 3, $I(h^+) + I(h^-) \leq I(f)$. Thus, $I(h^+), I(h^-) \leq I(f)$, and since f is integrable, $I(f)$ is finite. Therefore, both $I(h^+)$ and $I(h^-)$ are finite, and so h is integrable. **QED**

15. Let $f : \Omega \rightarrow \overline{\mathbb{R}}$ be a measurable function which is bounded on E , where $\mu(E) < \infty$, and $f = 0$ on E^c . Then, f is integrable.

Proof. We want to show that $I(f^+)$ and $I(f^-)$ are finite, but since both of them are bounded by $I(|f|)$, it suffices to show that $I(|f|)$ is finite.

Since f is bounded on E , $|f| < c$ for some $c \in \mathbb{R}$. Hence, $\int_E |f| d\mu \leq \int_E c d\mu = \int \chi_E c d\mu = c\mu(E) < \infty$, since $\chi_E c$ is a simple function. Further, $\int_{E^c} |f| d\mu = 0$, and so $\int |f| d\mu = \int_E |f| d\mu + \int_{E^c} |f| d\mu < \infty$, as required. **QED**